

Mini-Project 2

Group E

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1 Introduction

This mini-project involves the design and analysis of a 2D truss bridge using four fixed anchor points:

- Left outer: $(-1, 0)$
- Left inner: $(0, 0)$
- Right inner: $(20, 0)$
- Right outer: $(21, 0)$

The bridge must span a total length of 20 meters and support a vertical point load of 50 kN at the center of the span (node 5), located at a chosen height a . The goal is to perform both a static and a dynamic (modal) analysis for three different materials:

- Engineered wood,
- Aluminum alloy,
- Structural steel.

Project Objectives:

- Ensure the structure remains within the elastic range of the chosen material (no yielding).
- Limit the maximum displacement under the 50 kN load to 40 mm (L/500 criterion).
- Keep the total volume of each design under the specified limit:
 - Steel: 1 m^3
 - Aluminum: 1.7 m^3
 - Engineered wood: 2 m^3
- Avoid natural frequencies in the 1.5–10 Hz range, aiming for a first mode $> 50 \text{ Hz}$ where feasible.

Software and Approach: All analyses (static and dynamic) are carried out using MATLAB, by assembling global stiffness and mass matrices for the truss structure.

2 Methodology

2.1 Model Setup

In order to illustrate the finite element formulation of a truss, we begin with the classical one-dimensional axial bar problem in local coordinates, $x \in [0, L]$, which holds for each bar in the truss. We assume the bar has cross-sectional area A , Young's modulus E , and mass density ρ . Let $u(x', t)$ be the unknown axial displacement.

Strong Form. The strong (differential) form of the linear elastodynamic equation for the bar is given by

$$\begin{cases} E A \frac{\partial^2 u(x', t)}{\partial x'^2} = \rho A \frac{\partial^2 u(x', t)}{\partial t^2}, & x' \in (0, L), \\ E A \frac{\partial u}{\partial x'} \Big|_{x'=0} = -f_1(t), \\ E A \frac{\partial u}{\partial x'} \Big|_{x'=L} = f_2(t), \\ u(x', 0) = u_0(x'), \quad \frac{\partial u}{\partial t}(x', 0) = v_0(x'), \end{cases} \quad (1)$$

where $f_1(t)$ and $f_2(t)$ are external axial forces (no transverse loads or moments in truss structures) applied at the bar boundaries $x' = 0$ and $x' = L$.

From Strong to Weak Form. To derive the weak form (or variational formulation), we multiply the governing equation by a virtual displacement $\delta u(x')$, then integrate over $[0, L]$. For the left-hand side, we perform integration by parts:

$$u \in U \int_0^L \delta u \left(E A \frac{\partial^2 u}{\partial x'^2} \right) dx' = \left[E A \delta u \frac{\partial u}{\partial x'} \right]_0^L - \int_0^L E A \frac{\partial \delta u}{\partial x'} \frac{\partial u}{\partial x'} dx' \quad \forall \delta u \in V$$

The boundary terms $\left[E A \delta u \left(\frac{\partial u}{\partial x'} \right) \right]_0^L$, by applying the constitutive law and keeping the convention that forces are positive when pointing along the x' axis $E A \frac{\partial u}{\partial x'} = \pm f$, give rise to the end loads $f_1(t)$ and $f_2(t)$, thus yielding Neumann-Type Boundary Conditions. The resulting **weak form** becomes:

$$u \in U : \int_0^L E A \frac{\partial u}{\partial x'} \frac{\partial \delta u}{\partial x'} dx' + \int_0^L \rho A \frac{\partial^2 u}{\partial t^2} \delta u dx' = \delta u(0) [f_1(t)] + \delta u(L) [f_2(t)] \quad \forall \delta u \in V \quad (2)$$

This must hold for all virtual displacements u and δu in Sobolev Space H^1 :

$$H^1(\Omega) = \{u \in L^2(\Omega) \mid \nabla u \in L^2(\Omega)\}$$

with :

$$U = \{u \in H^1(\Omega) \mid u = 0 \text{ on } \Gamma\}$$

$V = \{v \in H^1(\Omega) \mid v = 0 \text{ at the ends of bars who connect to one of the four pin supports}\}$ (square integrable functions with square integrable first derivative), such that $\delta u = 0$ at the ends of bars who connect to one of the four pin supports. This could correspond to $x'=0$ or $x'=L$ depending on the local-global connectivity.

Semi-Discretization and Matrices. In a finite element approach, we approximate the local displacement $u(x')$ with shape functions $H_j(x')$ and nodal unknowns $q_j(t)$:

$$u_h(x') = \sum_j H_j(x') q_j(t), \quad \delta u_h(x') = \sum_j H_j(x') \delta q_j.$$

Plugging these into the weak form, one obtains a matrix equation for the local displacement of each bar.

$$\mathbf{M}_{\text{loc}} \ddot{\mathbf{q}}(t) + \mathbf{K}_{\text{loc}} \mathbf{q}(t) = \mathbf{F}_{\text{loc}}(t)$$

where:

- $\mathbf{M}_{\text{loc}}$ is the *mass matrix* (from the inertial term ρA) in local coordinates, $\mathbf{M}_{\text{loc},ij} = \int_0^L \rho A H_i^\top H_j dx$
- $\mathbf{K}_{\text{loc}}$ is the *stiffness matrix* (from the term $E A \partial u / \partial x ; \partial \delta u / \partial x$) in local coordinates, $\mathbf{K}_{\text{loc},ij} = \int_0^L E A \nabla H_i^\top \nabla H_j dx$
- $\mathbf{F}_{\text{loc}}(t)$ is the *forcing vector* (incorporating end loads f_1, f_2), $f_i = (f_2(t) + f_1(t)) H_i^\top(L)$
- $\mathbf{q}(t)$ is the vector of nodal displacements.

Element stiffness matrix in local coordinates:

$$K_{\text{loc}} = \int_0^\ell EA \begin{bmatrix} (h'_1)^2 & h'_1 h'_2 \\ h'_1 h'_2 & (h'_2)^2 \end{bmatrix} dx' = \frac{EA}{\ell} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Element consistent mass matrix in local coordinates:

$$M_{\text{loc}} = \int_0^\ell \rho A \begin{bmatrix} (h_1)^2 & h_1 h_2 \\ h_1 h_2 & (h_2)^2 \end{bmatrix} dx' = \frac{\rho A \ell}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Element applied loads vector in local coordinates:

$$f_{\text{loc}}(t) = \begin{bmatrix} h_1(0) \\ h_2(0) \end{bmatrix} f_1(t) + \begin{bmatrix} h_1(\ell) \\ h_2(\ell) \end{bmatrix} f_2(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \end{bmatrix}$$

Now, we consider the fact that bars are oriented by differing angles in a truss, and use this to write the local matrices in global x, y coordinates.

$$\begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{bmatrix} \begin{bmatrix} q_{1x} \\ q_{1y} \\ q_{2x} \\ q_{2y} \end{bmatrix}$$

Element stiffness matrix in global coordinates:

$$K = T^T K_{\text{loc}} T$$

$$= \frac{EA}{\ell} \begin{bmatrix} \cos^2(\theta) & \sin(\theta) \cos(\theta) & -\cos^2(\theta) & -\sin(\theta) \cos(\theta) \\ & \sin^2(\theta) & -\sin(\theta) \cos(\theta) & -\sin^2(\theta) \\ & & \cos^2(\theta) & \sin(\theta) \cos(\theta) \\ \text{Symm.} & & & \sin^2(\theta) \end{bmatrix}$$

Symm.

Element consistent mass matrix in global coordinates:

$$M = T^T M_{\text{loc}} T$$

$$= \frac{\rho A \ell}{6} \begin{bmatrix} 2 \cos^2(\theta) & 2 \sin(\theta) \cos(\theta) & \cos^2(\theta) & \sin(\theta) \cos(\theta) \\ & 2 \sin^2(\theta) & \sin(\theta) \cos(\theta) & \sin^2(\theta) \\ & & 2 \cos^2(\theta) & 2 \sin(\theta) \cos(\theta) \\ \text{Symm.} & & & 2 \sin^2(\theta) \end{bmatrix}$$

Regarding the forcing vector, it is much more intuitive to write the global forcing vector after assembling global mass and stiffness matrix rather than attempting to assemble local forcing vectors.

We now perform the assembly operation in order to produce global mass and stiffness matrices. This is done by initialising square matrices with size equal to the global number of degrees of freedom (twice the number of nodes). The connectivity table is then used to add element mass or stiffness contributions to the corresponding global node number in the matrix.

We consider a two-dimensional truss structure, composed of multiple bar (rod) elements arranged in a triangulated fashion, spanning 20 m with additional supports at coordinates $(-1, 0)$, $(0, 0)$, $(20, 0)$, and $(21, 0)$. Each of these **anchor nodes** is fully pinned, meaning the in-plane displacements satisfy

$$u_x = 0, \quad u_y = 0 \quad \text{at the anchor supports.}$$

A concentrated downward load $F = 50 \text{ kN}$ is applied to the **midspan node**, located at $(10, a)$. All other nodes are free to translate in the global x - y plane, subject only to internal axial forces in the truss members.

We apply this pinned boundary conditions by simply deleting rows and columns of the mass and stiffness matrices where displacement is set to zero by the pinned supports.

The global forcing vector will simply be a vector containing zeros everywhere except - 50kN at the y-degree of freedom at the node where the force is applied.

Consequently, the semi-discrete global problem has been generated, a matrix equation with dimension equal to the global number of free degrees of freedom.

$$\mathbf{M} \ddot{\mathbf{q}}(t) + \mathbf{K} \mathbf{q}(t) = \mathbf{F}(t),$$

Geometry and Nodes:

We model the truss in a 2D plane with the following constraints:

- Four anchor points at the given coordinates, fully fixed (pinned in 2D, displacements $u_x = 0, u_y = 0$).
- The midspan node is at $(10, a)$, where a is chosen freely.
- Additional intermediate nodes and linked with bars to create the bridge's truss structure.

2.2 Material Properties

We consider three separate designs, each using a distinct material:

Material	E (GPa)	ρ (kg/m ³)	Cross-Sectional Area
Structural steel	200	7850	15–50 cm ²
Aluminum alloy	69	2700	20–70 cm ²
Engineered wood	12	600	30–100 cm ²

For each material, the area of each truss member is optimized so that:

- The total volume does not exceed the limit specified (1 m³, 1.7 m³, or 2 m³).
- Maximum von Mises stress (or axial stress in a truss) remains below yield stress (thus preserving elasticity).
- Vertical displacements under the 50 kN load remain below 40 mm.

2.3 Boundary Conditions and Loads

- **Supports:** Nodes at $(-1, 0)$, $(0, 0)$, $(20, 0)$, and $(21, 0)$ have $u_x = 0$, $u_y = 0$.
- **Load:** A concentrated downward force of 50 kN is applied at node 5 (i.e., $(10, a)$).

2.4 Analysis Procedures

Static Analysis:

- Assemble the global stiffness matrix K .
- Apply boundary conditions and known nodal loads.
- Solve $K\mathbf{u} = \mathbf{F}$ for the displacement vector $\mathbf{u}$.
- Compute element axial stresses $\sigma = \frac{E}{L}(u_{\text{end2}} - u_{\text{end1}})$ in local coordinates.
- Check displacements and stresses against requirements.

Modal Analysis:

- Assemble the global mass matrix M by summation of each truss element's mass contribution.
- Solve the eigenvalue problem $[K - \omega^2 M]\phi = \mathbf{0}$.
- Extract the first natural frequency $\omega_i = 2\pi f_i$ and mode shapes ϕ_i .
- Check that f_1 is not within 1.5–10 Hz, and ideally $f_1 > 50$ Hz if feasible.

2.5 Assumptions and Simplifications

- Perfect pin-jointed connections (no bending moments).
- Linear elastic material behavior (no plasticity).
- Small displacements (geometric nonlinearity ignored).
- No damping considered for the modal analysis.
- Neglect the influence of gravitation self-weight of the bars.
- Small truss deformations.

3 Truss Optimization Methodology and Results

3.1 Wooden Warren Truss

The first task is to produce a truss design which satisfies displacement, maximal stress and natural frequency criteria for an engineered wood material. The following Warren Truss is considered as a starting point.

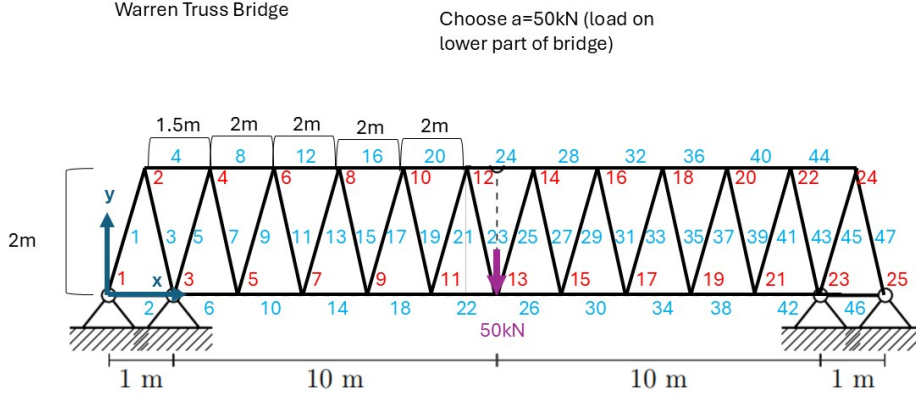


Figure 1: Warren Truss Bridge

Table 1: Warren Truss Analysis Results

Iter	Material Volume m^3	Max Nodal Disp (mm)	1st Nat Freq (Hz)	Max Stress (MPa)
1	0.9597	23.1	23.8	10.3
2	0.3654	49.5	29.7	14.2
3	0.5048	39.5	25.8	14.3
4	0.5398	34.8	27.4	10.9

For iteration 1, all bars are assigned their maximum allowable cross-sectional area (100 cm^2). Project2.mlx is run to perform modal and static analysis.

This first design results in a maximum nodal displacement about half the allowable amount in node 13 and a yield safety factor of 2.9 in bar 24. The first natural frequency is well outside of the forbidden 1.5-10Hz range, but could be pushed up to hopefully surpass the desirable threshold of 50Hz. Material volume is half the maximum limit.

For the second iteration, we notice that the safety factor is quite conservative and aim to lower it to 2. We begin by thinning out all lower stress bars to 30 cm^2 in order to save on material volume. Bars with a safety factor less than 10 (yellow category) are set to 50 while safety factors less than 6 (red category) are set to 75.

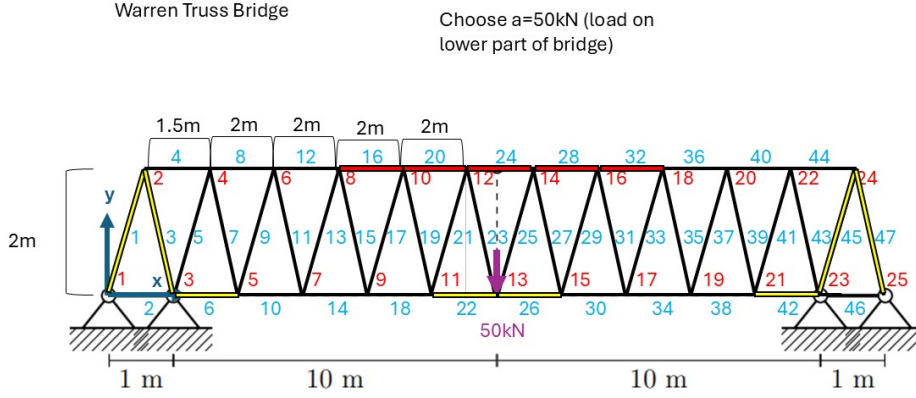


Figure 2: Warren Truss Bridge Iteration 1 with High Load Members Shown in Red and Yellow

The second iteration cut material use by nearly 3, but led to violating the maximum displacement criterion. For iteration 3, noncritical bars will be increased to 50cm^2 , thus matching those in the yellow category, to rigidify the structure. Iteration 3's design addressed the displacement limit without significant changes to the maximum stress. The remaining objective is to attempt to increase the first natural frequency. This first mode corresponds to a sinusoidal bending profile with wavelength equal to the bridge span and is shown in figure 3.

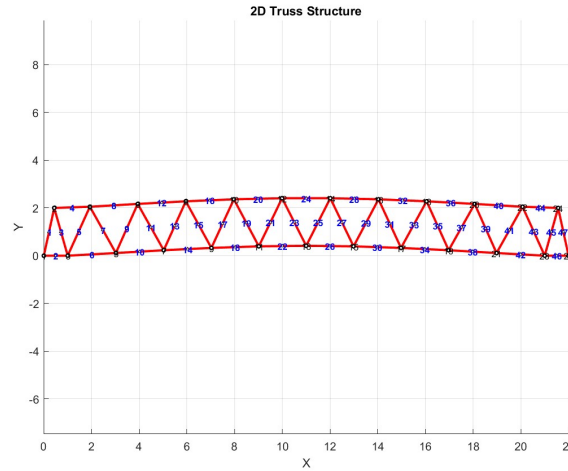


Figure 3: First Mode for Bridge at Iteration 3

We compute the elongation of bars under mode 1 and find that bars 6, 20, 24, 28 and 42 are the most deformed under mode 1. By enlarging and thus rigidifying them, the frequency of mode 1 can hopefully be increased. These five bars are increased to 100cm^2

for iteration 4.

Ultimately, this heuristic approach is only moderately successful in increasing the fundamental natural frequency as deformation of the mode is transferred to other bars and little stiffening effect is achieved. Yet, thanks to its lowering of nodal displacement and maximum stress for a very moderate increase in material use, iteration 4 can be considered to provide a fairly optimal bridge design. Significant increases in natural frequency can likely be achieved by changing node coordinates, switching to an entirely different bridge design and considering other materials, which will be explored in the next bridges.

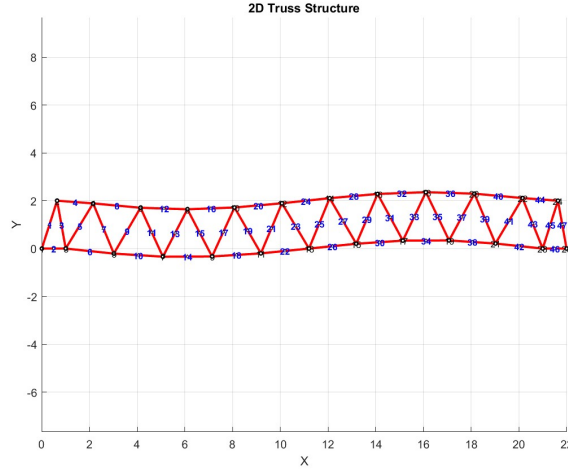


Figure 4: Second Mode for Bridge at Iteration 4

As an extension, bars under compression are checked for risks of individual bar buckling. This requires making a specific choice regarding bar section geometry: an annular section with inner radius 0.9 times the outer radius is taken. The minimum safety factor is 16.1 under the 50kN static load, a sufficient safety to handle perturbations and transient load effects. Further linear perturbation analysis of the entire structure would be necessary to ensure no global buckling of the structure (rather than buckling involving only individual bars) is possible.

3.2 Aluminium Camelback Truss

Attempts to increase the first natural frequency by modifying specific bar cross-sections had limited success. While increasing all bar cross-sections would stiffen the bridge, this approach is costly in terms of material volume, which is also an optimization objective. This motivates the consideration of a different bridge geometry, with a wider truss in the center, to hopefully stiffen this fundamental mode while adding less material volume.

We therefore select the Camelback Truss to inspire this bridge design. With a greater height in the center, the truss is effective at reducing central displacement and stiffening the bridge compared to the Warren Truss.

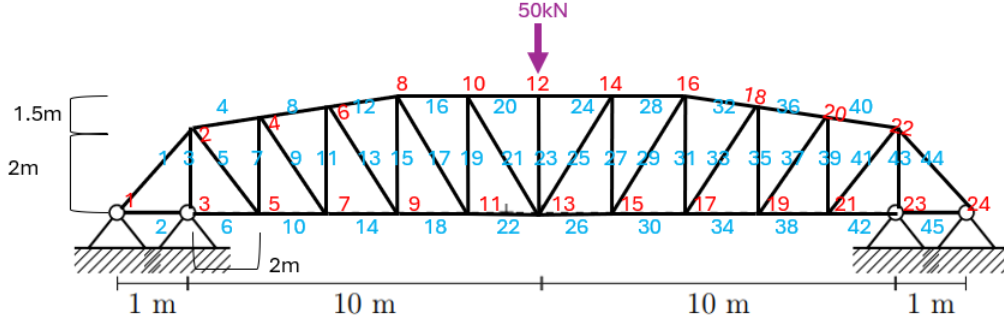


Figure 5: Camelback Truss Bridge

Table 2: Camelback Truss Analysis Results

Iter	Material Volume m^3	Max Nodal Disp (mm)	1st Nat Freq (Hz)	Max Stress (MPa)
1	0.8032	4.1	25.6	9.94
2	0.2491	11.5	27.8	19.0
3	0.2864	10.2	30.7	18.1
4	0.3056	9.0	34.3	17.9

In iteration 1, we run the analysis with maximum bar thickness for all bars. Then, similarly to the Warren Truss, we reduce bar thicknesses according to stress safety factors. Safety factors below 25 are set to 40 cm^2 , below 35 set to 30 and all others set to the minimal 20 cm^2 .

With these changes, iteration 2 performs very well on material volume, stress and displacement criteria. To increase natural frequency, we increase thickness of the most deformed bars (under this mode) to 50.

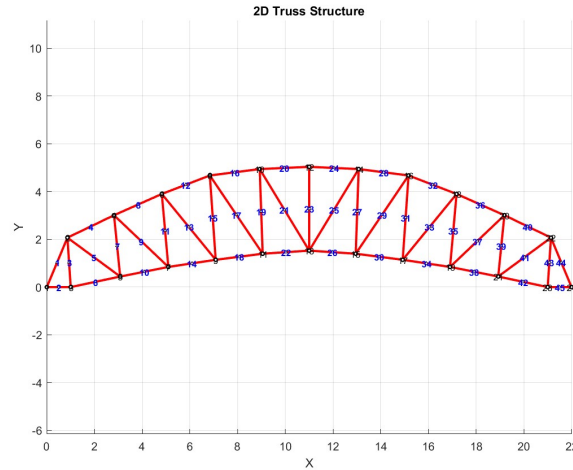


Figure 6: Truss Iteration 3 Fundamental Mode

Since, once again, this approach only achieves modes increases in fundamental frequency, for iteration 4 we will try adding bars. One notices that the diagonal bars in the Camelback truss are all directed towards the center. Bars along the other diagonal

would be highly stretched by the fundamental mode (see figure). To rigidify the truss, we therefore add two reverse diagonal bars near the supports for iteration 4, a region which bends significantly under both the first mode and static loading, thus rigidifying the bridge.

The approach is quite successful as iteration 4's fundamental frequency is increased to 34.3 Hz for marginal increases in material volume.

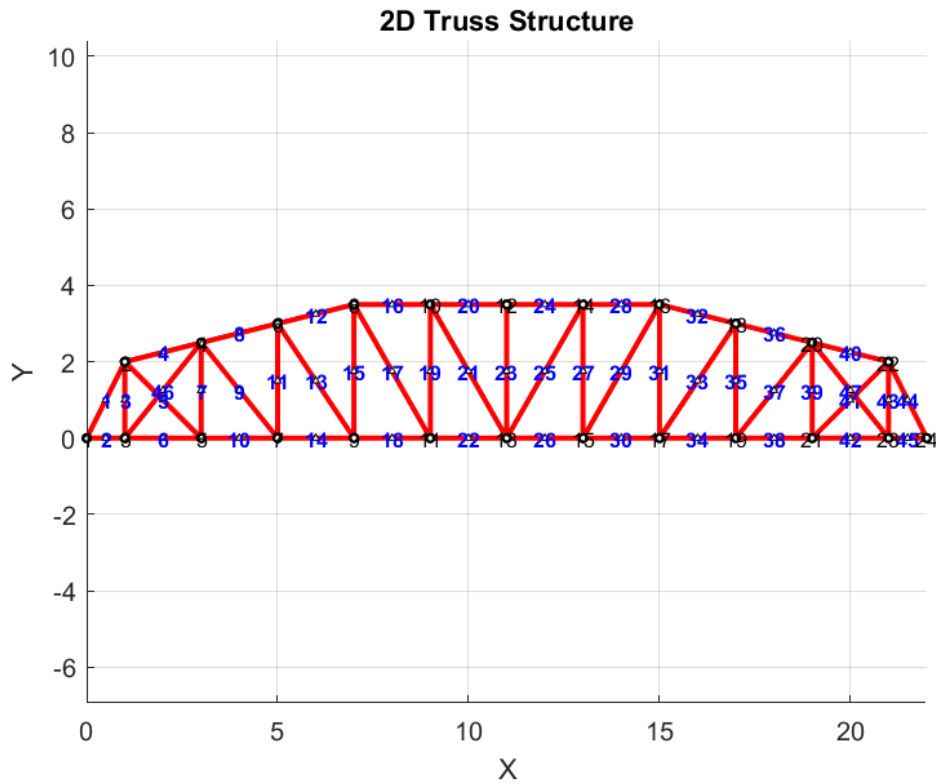


Figure 7: Modified Camelback with 2 Extra Bars

At this final iteration, members under compression are verified for risks of individual bar buckling, considering the same annular section as before. The minimum safety factor is 6.7 under the 50kN static load at bars 19 and 27, a smaller but sufficient safety to handle perturbations and transient load effects. If an additional margin is deemed necessary, these two members could be simply be thickened. Again, further linear perturbation analysis of the entire structure would be necessary to ensure no global buckling of the structure is possible.

3.3 Metal Pratt Truss

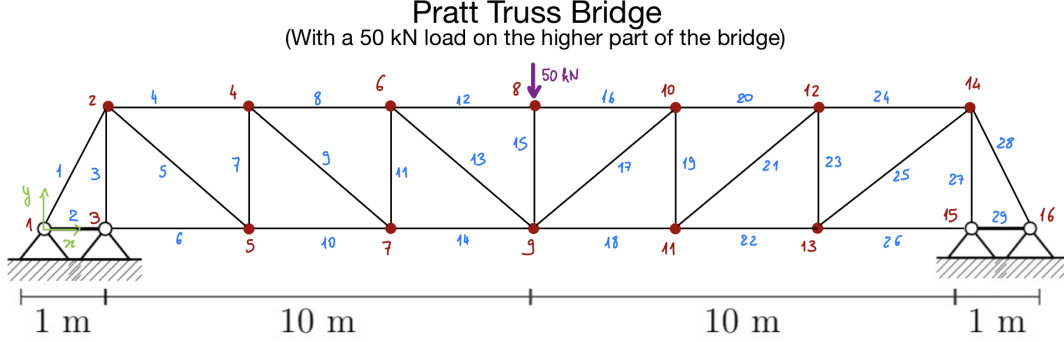


Figure 8: Pratt Truss Bridge

This section presents and interprets the results obtained for the third bridge, a steel Pratt Truss. The analysis covers the distribution of axial stresses, deflections under the 50 kN load, verification of the design against yielding or buckling, and a brief modal analysis to evaluate natural frequencies.

Under the 50 kN concentrated load applied at mid-span ($x = 10 \text{ m}$), the following observations were made:

- **Diagonals in tension:** In a Pratt configuration, the diagonal members are primarily in tension, which aligns well with the design philosophy to minimize buckling risks. The maximum axial stresses in these diagonals remain within the elastic limit of the steel, providing a comfortable safety margin (e.g. factor of safety between 1.5 and 2, depending on final cross-sectional areas).
- **Verticals in compression:** The vertical members experience moderate compression. Overall, their stress utilization stays below about 80% of the allowable limit, and no critical buckling issues arise, assuming appropriate sections are chosen to keep slenderness under control.

Hence, the Pratt layout channels the main tension forces through its diagonals and avoids excessive compressive loading on slender elements.

The maximum vertical deflection at the mid-span node (node 8) under the 50 kN load is approximately $\delta_{\max} \approx 3 \text{ mm}$ (from the finite element analysis). This value remains far below the specified service limit of 40 mm, thus meeting displacement criteria.

Table 3: Pratt Truss Analysis Results

Iter	Material Volume m^3	Max Nodal Disp (mm)	1st Nat Freq (Hz)	Max Stress (MPa)
1	0.4190	1.95	29.43	16.4
2	0.3352	2.43	29.43	20.4
3	0.2514	2.96	29.43	23.6
4	0.2976	2.92	28.07	27.2

Compared to the first two bridges (e.g. Warren or Camelback truss), the Pratt design might exhibit slightly different deflection values due to its diagonal arrangement and selected cross-sectional areas. Also, we have used fewer nodes and elements.

By optimizing the cross-sectional areas within the specified range ($15\text{--}50\text{ cm}^2$), the total steel volume is approximately 0.3 m^3 . We tried to optimize the total volume : some bars require larger sections (e.g. the main diagonals), whereas others that carry less force can be reduced to save material and overall weight. This results in iteration 4, the first and second modes are plotted in Figure 10.

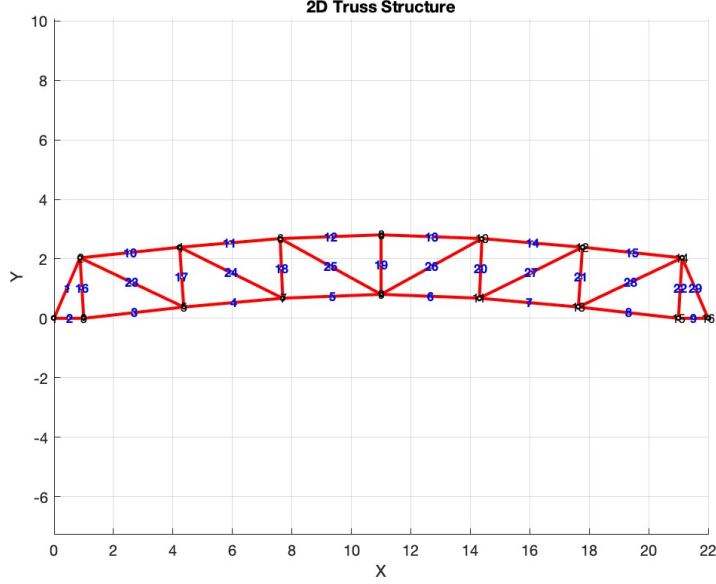


Figure 9: First Mode for Pratt Bridge at Iteration 1

This first mode corresponds to a sinusoidal bending profile with wavelength equal to the bridge span and is shown in figure 9. We computed it in iteration 1 with a cross-sectional area of 50 cm^2 .

Though the study focuses primarily on static behavior, a simple modal analysis was performed by solving $(\mathbf{K} - \omega^2\mathbf{M})\boldsymbol{\phi} = 0$. The first few natural frequencies, ω_i , are found to be above 25 Hz, keeping the bridge away from common resonance ranges (1.5–10 Hz). The overall balance between mass and stiffness leads to reasonable mode shapes with no large amplitude deformations near the mid-span under typical dynamic excitations.

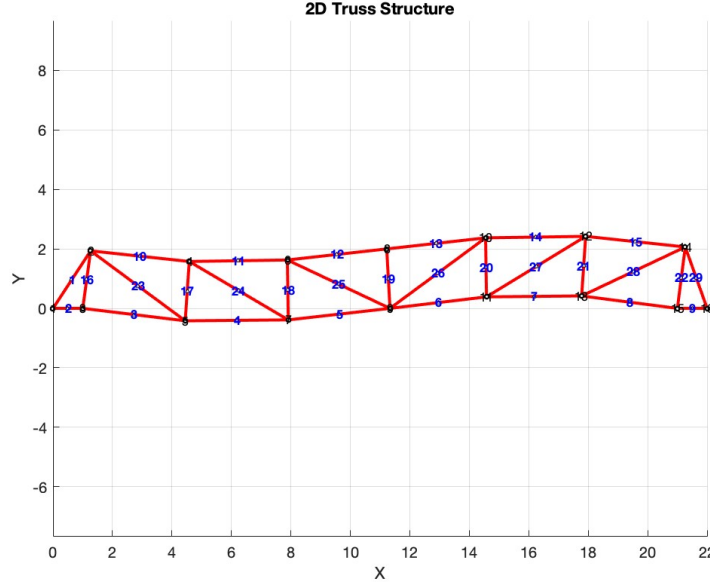


Figure 10: First and Second Mode for Pratt Bridge at Iteration 4

- **Mechanical performance:** Axial forces are effectively directed into tension in the diagonal members, reducing buckling concerns. Compressive elements remain well within allowable limits.
- **Buckling:** The risk of buckling for the truss members under compression depends on the critical buckling load, which can be calculated using Euler's formula for a column under axial compression:

$$P_{cr} = \frac{\pi^2 EI_x}{L^2}$$

Where:

- P_{cr} is the critical buckling load,
- E is the Young's modulus of the material,
- I_x is the moment of inertia of the cross-section about the x -axis,
- L is the length of the member.

If we assume a circular cross-section, the moment of inertia I_x is given by:

$$I_x = \frac{\pi r^4}{4}$$

Where r is the radius of the section. Using the provided material properties and dimensions, the critical buckling load for the members is calculated to be approximately:

$$P_{cr} \approx 982 \text{ kN}$$

This means that if the applied load on any member exceeds 982 kN, the member is at risk of buckling. If the applied load is less than this value, the members will remain stable under compression. The maximum force on each member (maximum

stress divided by area of the cross section), calculated based on the given maximum stress of 27 MPa, is 135 kN. This is significantly lower than the critical buckling load of 982 kN. Therefore, there is no risk of buckling as long as the applied force remains below this value.

- **Deflection:** The maximum mid-span deflection under 50 kN is safely below the 40 mm requirement, indicating good serviceability.
- **Weight vs. Strength:** Despite the more intricate geometry, properly chosen cross-sections keep total volume in a similar range to the other designs. A prudent safety margin is still maintained.

Compared to the previous two bridges, the *Pratt Truss* offers a favorable distribution of forces, with the main diagonals in tension. This approach leads to reduced risk of compression-related failures and achieves acceptable stiffness for a 20 m span under a 50 kN load.

Overall, the steel Pratt Truss proves to be an efficient and robust option, limiting mid-span deflection, keeping all axial stresses below yield, and maintaining an acceptable total mass. Its capacity to direct tension along diagonal members helps mitigate buckling issues and provides a balanced structural solution for this span length and load scenario.

4 Conclusion

The optimization process has resulted in 3 distinct Warren, Camelback and Pratt-inspired bridge truss designs. At their final iterations, all designs respected criteria the 0.04m maximum displacement limit, safety factor of 2 in stress, material volume limits and natural frequencies were kept over 15 Hz. None of the 3 bridge designs met the more ambitious 50 Hz target, which would likely require the use of smaller bars to form a denser, stiffer truss. An extension would be to perform linear perturbation analysis on the entire truss structure as critical buckling modes could involve multiple bars deflecting. Considering the self-weight of bars in the static loading calculations (the aluminium camelback bridge weighs 8kN) would also be a useful extension given this non-negligible conclusion. However for a rough estimate, placing this self-weight load at the central node could provide a safe bound for design.

5 Appendix

- MATLAB code for FEM implementation.